

Survey

Version

Front axle 435.110/111/113/160/170

33.3

Front axle 435.115/117

33.6

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Installation Survey

Chassis		Front axle		Brake	Installation
Model	Sales designation	Model	Sales designation		
435.110	U 1700	737.200 ¹⁾	AU 3/1 S – 5,3	Disk brake	Series
.111	U 1700 L	.202 ²⁾			
.113	U 1700 L/38				
.160	Kit TM 170	737.232	AU 3/1 S – 6,8		
.170	Kit Condor	737.233	AU 3/1 S – 6,2		
435.111	U 1700 L	737.201	AU 3/1 S – 5,3		SA

1) Valid to chassis end No. 084 388

2) Valid from chassis end No. 084 389

Overall View



UR 33-0339

Front axle

33.3 General

737.2

Technical Data

Ratios

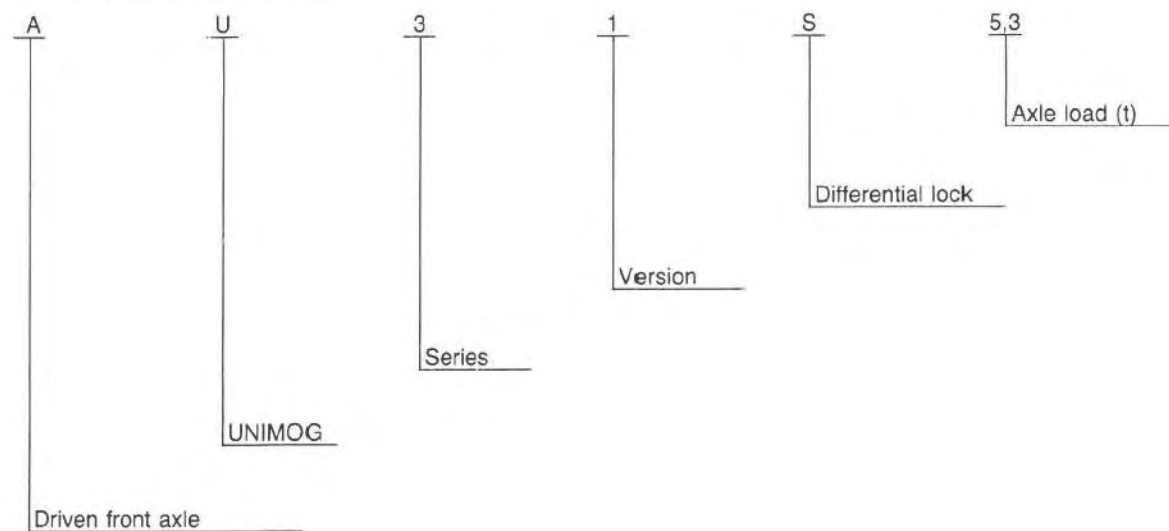
Axle model	Installation	Gear train Ring gear drive pinion		Gear train Wheel hub drive		Total Ratio
		Number of teeth	Ratio	Number of teeth	Ratio	
737.200 .202 .231/.233 .232	Series	24: 11	2,18	38: 13	2,92	6,38
737.201	SA	24: 13	1,85			5,40
737.202	SA ¹⁾	25: 9	2,78			8,12

1) Only applies to Model 435.110.

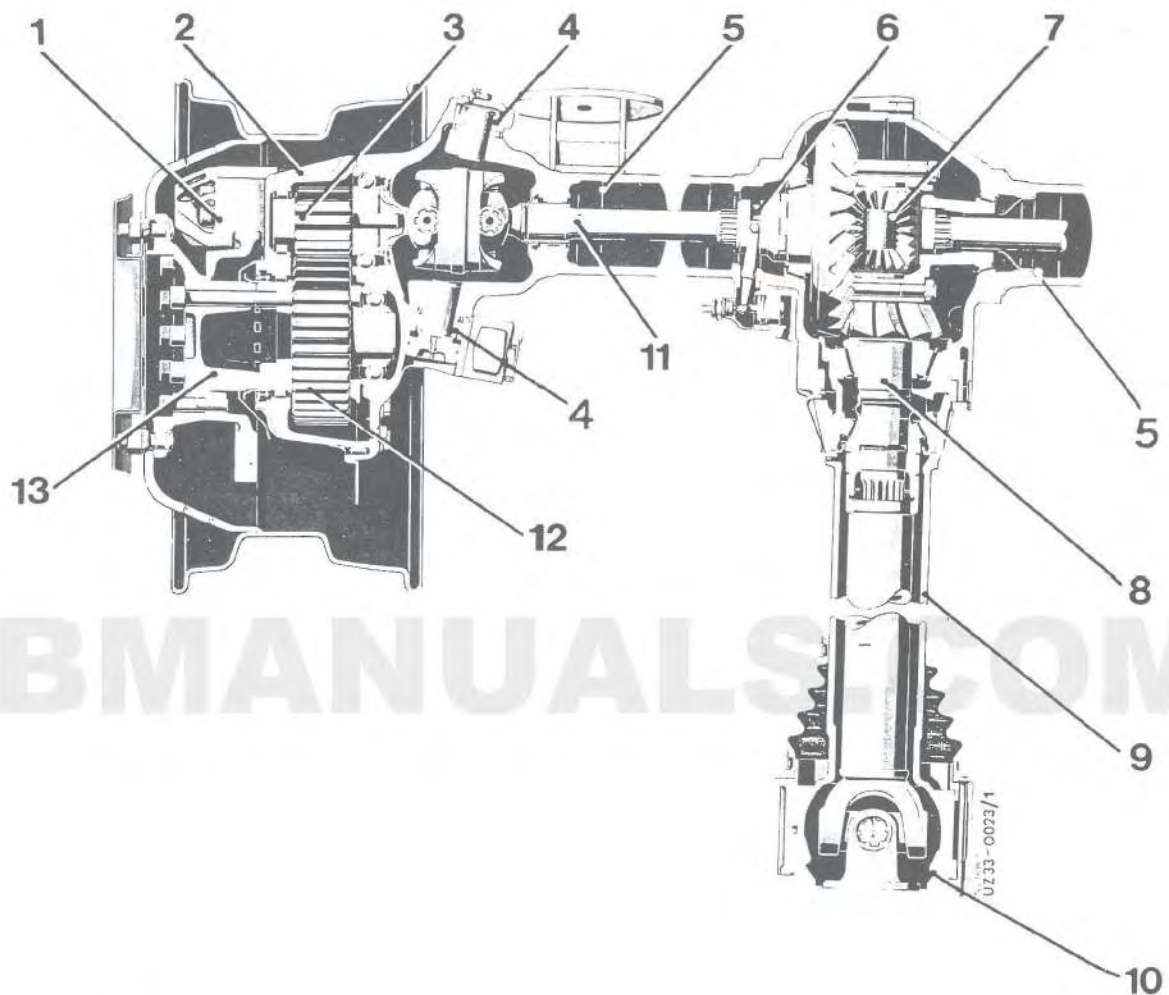
Capacities

Designation	Service product	SAE class	Capacity (L)
Axle center housing	Hypoid gear oil	90	2,5
Wheel hub drive			each 0.6

Key to Front Axle Designation



Sectional View



AU 3/1S-5.3

- | | | | |
|---|-------------------|----|-------------------------------|
| 1 | Fixed caliper | 8 | Drive pinion |
| 2 | Wheel hub drive | 9 | Torque tube |
| 3 | Drive gear | 10 | Torque ball casing |
| 4 | Swivel joint | 11 | Constant-velocity joint shaft |
| 5 | Axle tube | 12 | Driven gear |
| 6 | Differential lock | 13 | Wheel hub |
| 7 | Differential | | |

Setting Values

Designation	Setting Values
Preload of torque ball shells on extension tube mounting flange manual effort measured	200 to 300 N
Axle drive housing to large tapered roller bearing, lock end	± 0
Axle drive housing to small tapered roller bearing Preload (see axle data sheet)	0,2 mm
Axle tube to pinion flange	± 0
Deep-groove ball bearing to driven gear in wheel hub drive	± 0
Breakaway force for drive pinion, (see axle data sheet)	6,0 to 6,5 nm
Distance (A) from the frame side member to center of hole on pitman arm	165 mm
Preload, steering knuckle bearing	0,10 to 0,15 mm

Expendable Materials

Cons. No.	Designation	Part Number
1	Teroson Atmosit sealing compound	000 989 43 20 ¹⁾
2	Long-term grease	commercially available
3	Molybdenum disulphide running-in paste	commercially available
4	Loctite No. 241 adhesive	000 989 70 71
5	Loctite No. 270 adhesive	002 989 93 71
6	Molykote liquid grease	commercially available
7	Molykote paste D X	commercially available
8	Hypoid gear oil SAE 90	commercially available
9	Terostat 56 surface sealing compound	001 989 58 20

1) optional 001 989 58 20

General note: In order to ensure satisfactory adhesion of sealing compounds, the sealing surfaces must be completely free of oil and grease.

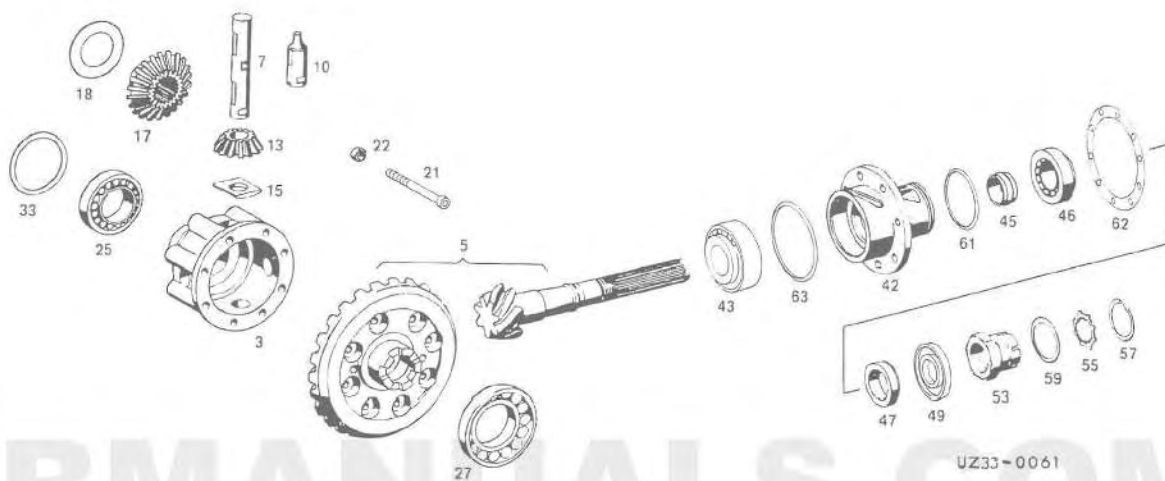
Special Tools

Cons. No.	Tool	Number	Tool set
1	Extension tube for mounting flange	000 589 01 14 00	A
2	Allan key socket 14 mm, 3/4" square for brake disk	001 589 68 09 00	B
3	Allan key socket 17 mm, 3/4" square for wheel hub	001 589 69 09 00	B
4	Remover and installer for driven gear	425 589 04 14 00	B
5	Guide pin (2 required) for wheel hub	425 589 01 15 00	B
6	Guide pin (3 required) for wheel hub drive housing	425 589 02 15 00	B
7	Drift for installer	406 589 04 15 00	B
8	Puller for drag link and track rod	406 589 05 33 00	B
9	Extractor for swivel joint	425 589 00 33 00	B
10	Installer (4 required) for driven gear	425 589 02 33 00	B
11	Counter-support for pullers and extractors	000 589 35 33 00	B
12	Extractor for brake pads	001 589 43 33 00 (115 589 14 33 00)	B
13	Puller for counter-support (with tool No. 11)	425 589 00 34 00	B
14	Installer for deep-groove ball bearing and shaft seal in wheel hub drive (with tool No. 7)	425 589 00 43 00	B
15	Installer for shaft seal in steering knuckle (with tool No. 7)	425 589 03 43 00	B
16	Installer for deep-groove ball bearing and shaft seal in axle tube (with tool No. 7)	425 589 05 43 00	B
17	Assembly flange for torque ball casing (with tool No. 1)	425 589 00 59 00	B
18	Torque wrench for drive pinion (4 to 6 Nm, 1/4" square)	000 589 67 21 00	C
19	Adjuster for drive pinion (with tool No. 18)	425 589 00 21 00 (406 589 10 21 00)	C
20	Mounting for drive pinion	425 589 00 31 00 (406 589 01 31 00)	C
21	Assembly sleeve for taper roller bearing in drive pinion	425 589 00 35 00	C
22	Assembly sleeve for swivel joint	425 589 01 35 00	C
23	Installer for roller bearing in wheel hub drive housing (with tool No. 18)	425 589 01 43 00	C
24	Installer for taper roller bearing in axle tube (with tool No. 7)	425 589 02 43 00	C
25	Installer for shaft seal in drive pinion (with tool No. 21)	425 589 04 43 00	C

Tightening Torques

Designation	Thread	Nm
Wheel hub to drive gear	M 20 x 1,5	600
Drive gear to drive shaft	M 20 x 1,5	600
Fixed caliper to axle	M 20 x 1,5	500
Track rod arm/steering knuckle arm to steering knuckle	M 18 x 1,5	400
Wheel nut	M 22 x 1,5	400
Shock absorber mounting to axle	M 20 x 1,5	400
Brake disk to wheel hub	M 18 x 1,5	400
Axle strut to axle tube	M 18 x 1,5	350
Axle strut to torque tube	M 18 x 1,5	350
Front spring to spring bracket	M 18 x 1,5	300
Transverse link to frame	M 16 x 1,5	300
Differential assembly	M 16 x 1,5	300
Track rod mounting	M 20 x 1,5	230
Stop to steering knuckle, inner	M 16	200
Axle tube assembly	M 14 x 1,5	200
Wheel hub drive housing to steering knuckle	M 14 x 1,5	200
Torque tube to pinion flange	M 14 x 1,5	200
Propeller shaft to transmission	M 12 x 1	100
Clamp to track rod	M 12 x 1,5	100

Exploded View

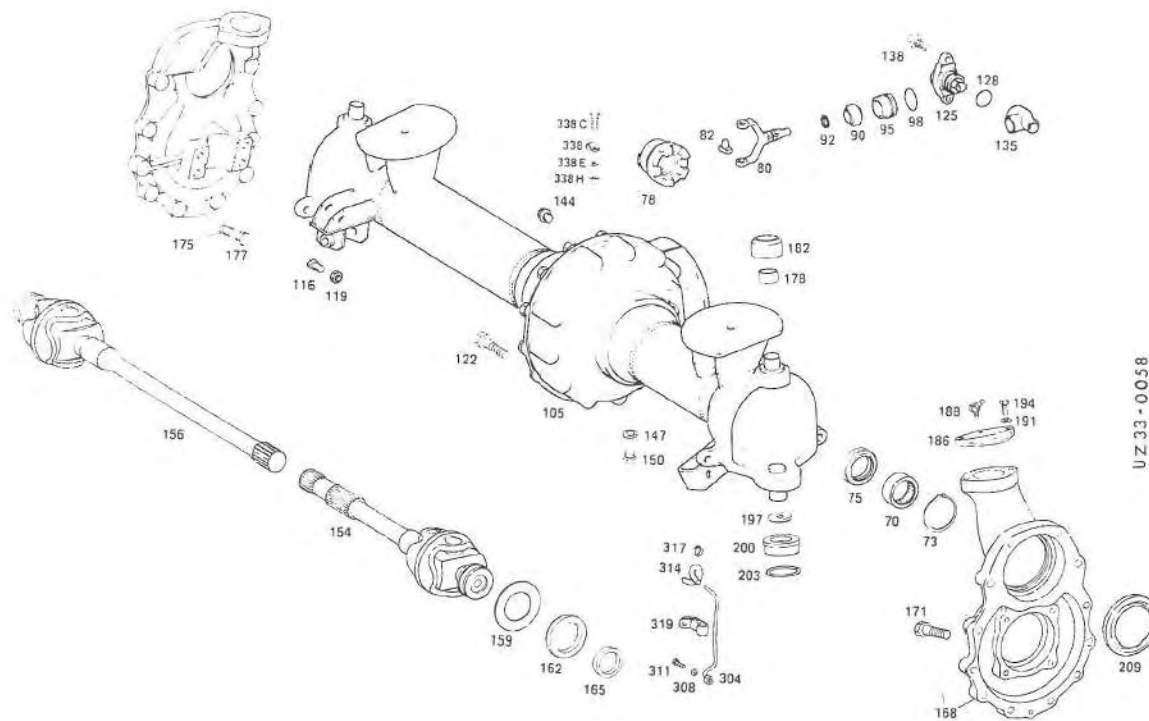


Differential housing and drive pinion

3	Housing	42	Pinion flange
5	Gear set	43	Taper roller bearing
7	Differential pinion shaft	45	Spacing sleeve
10	Differential pinion shaft	46	Taper roller bearing
13	Bevel gear	47	Wear ring
15	Thrust washer	49	Sealing ring
17	Side gear	53	Nut
18	Thrust washer	55	Retainer
21	Bolt	57	Snap ring
22	Nut	59	O-ring
25	Taper roller bearing	61	O-ring
27	Taper roller bearing	62	Spacer
33	Shim	63	O-ring

737.2

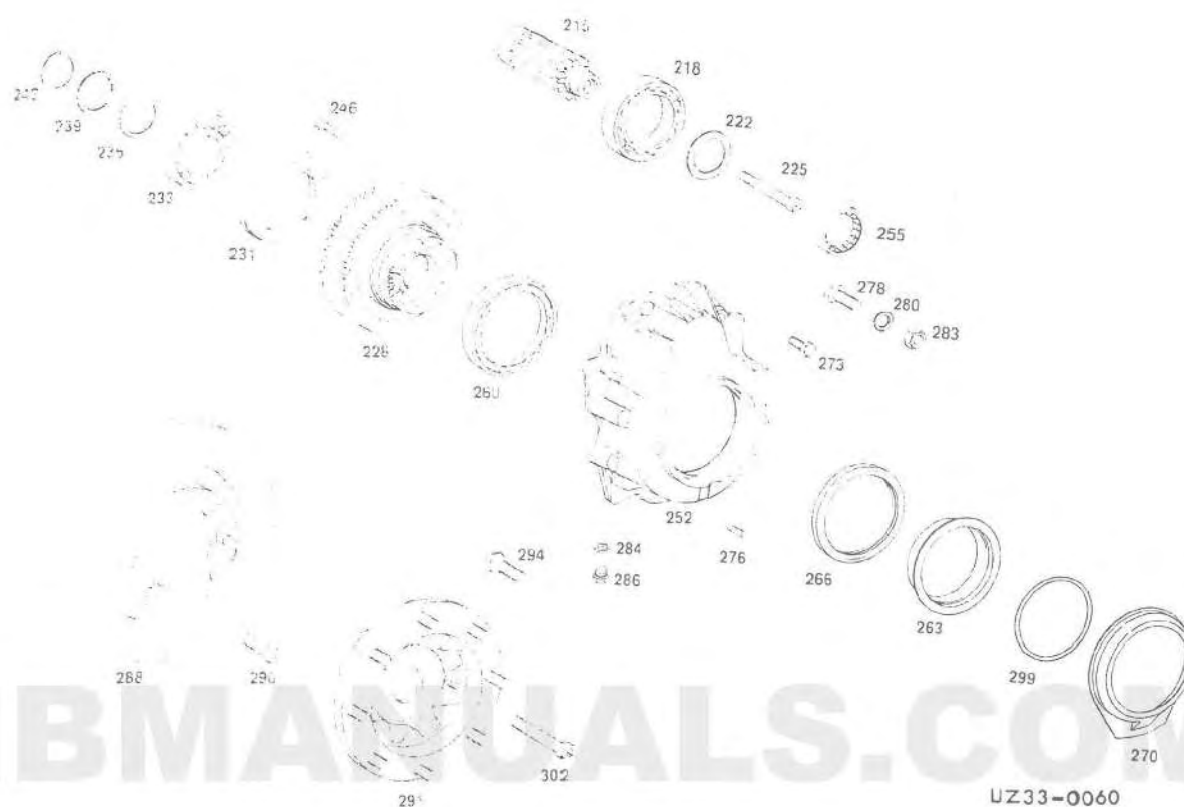
Exploded View



Axle tube and steering knuckle

70	Needle bearing	165	Sealing ring
73	Snap ring	168	Steering knuckle
75	Sealing ring	171	Bolt
78	Shift dog	175	Steering knuckle
80	Selector fork	177	Bolt
82	Sliding lock	178	Bushing
90	Swivel joint	182	Swivel joint
92	Snap ring	186	Plate
95	Bushing	188	Grease nipple
98	Sealing ring	191	Spring washer
105	Axle tube	194	Bolt
116	Bolt	197	Thrust washer
119	Nut	200	Thrust shell
122	Bolt	203	Sealing ring
125	Shift cylinder	209	Sealing ring
128	Sealing ring	304	Line
135	Housing	308	Sealing ring
138	Bolt	311	Union screw
141	Switch	314	Hose
144	Screw plug	317	Clamp
147	Sealing ring	319	Clamp
150	Screw plug	338	Clamp
154	Drive shaft	338C	Bolt
156	Drive shaft	338E	Lock washer
159	Shim	338H	Nut
162	Wear ring		

Exploded View

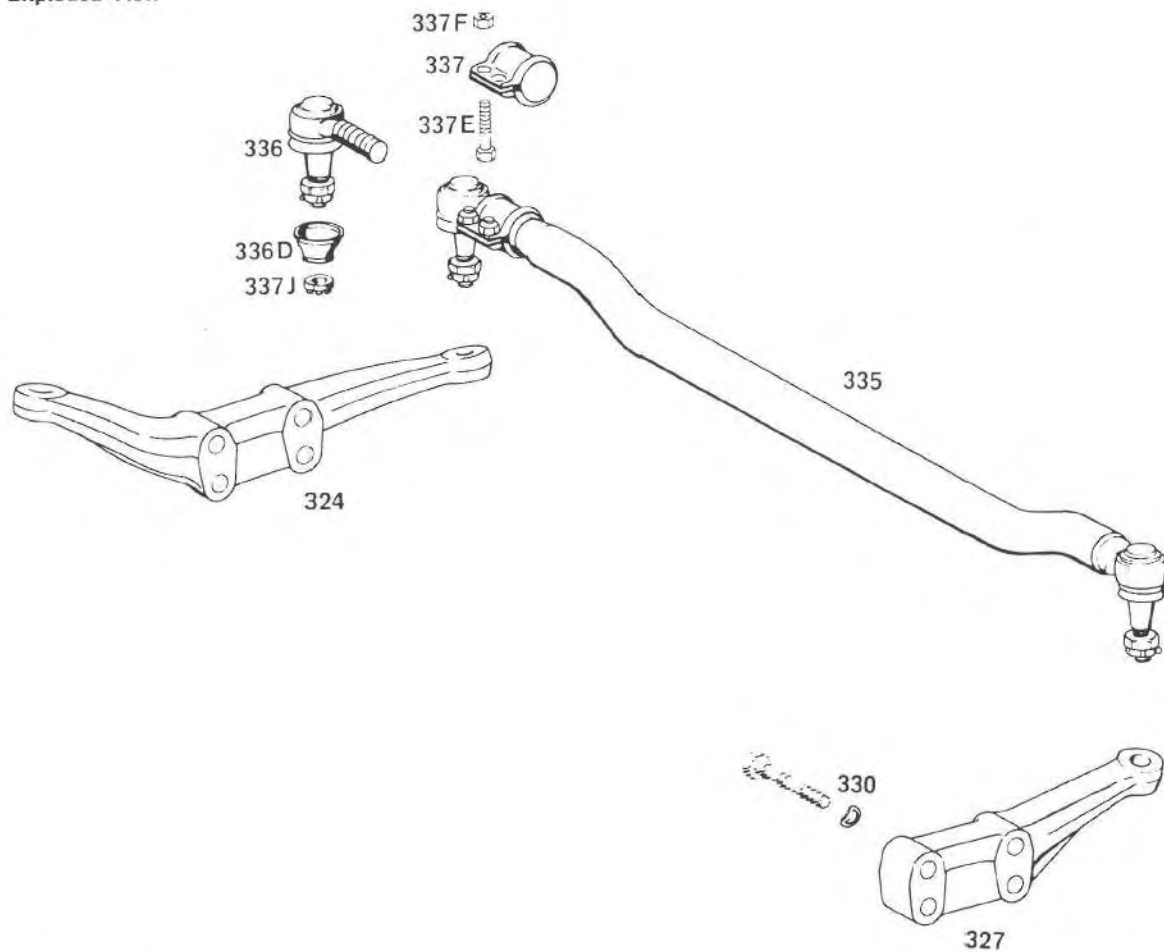


Wheel hub drive

215	Drive gear	266	Sealing ring
218	Ball bearing	270	Oil baffle
222	Shim	273	Bolt
225	Bolt	276	Straight pin
228	Driven gear	278	Bolt
231	Bearing ring	280	Lock washer
233	Ball bearing	283	Nut
236	Shim	284	Sealing ring
239	Shim	286	Screw plug
243	Snap ring	288	Brake disk
246	Bolt	291	Wheel hub
252	Housing	294	Wheel stud
255	Cylindrical roller bearing	296	Bolt
260	Cylindrical roller bearing	299	O-ring
263	Wear ring	302	Bolt

737.2

Exploded View



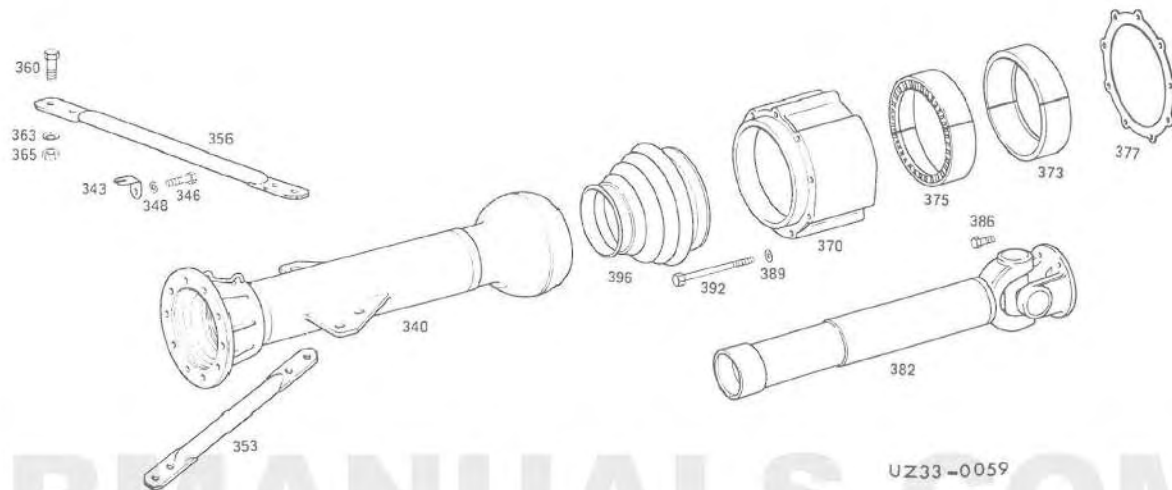
UZ46-0038

Steering components

324 Steering knuckle arm
 327 Steering arm
 330 Spring washer
 335 Track rod
 336 Ball joint end

336D Collar
 337 Clamp
 337E Bolt
 337F Nut
 337J Nut

Exploded View

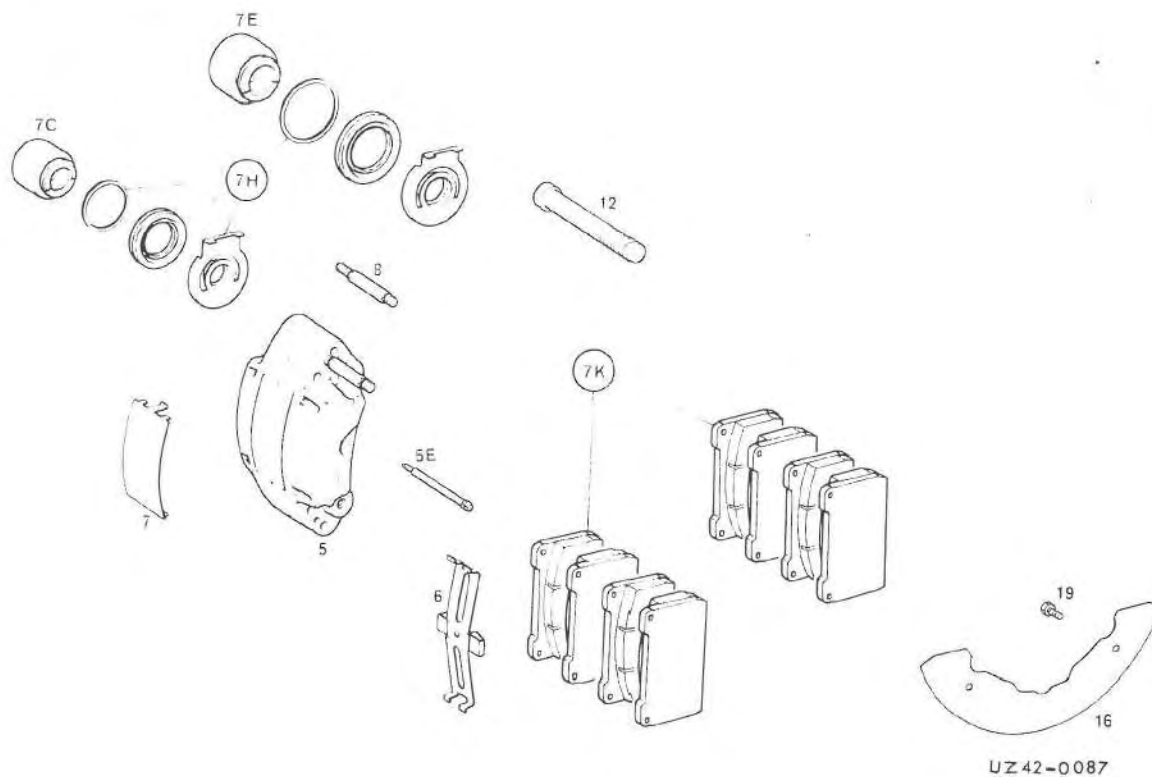


Thrust tube

340	Thrust tube	370	Torque ball casing
343	Bracket	373	Torque ball shell
346	Bolt	375	Torque ball shell half
348	Spring washer	377	Shim
353	Axle strut	383	Propeller shaft
356	Axle strut	386	Bolt
360	Bolt	389	Sealing ring
363	Conical spring washer	392	Bolt
365	Nut	396	Rubber gaiter

737.2

Exploded View



Brake components

5	Fixed caliper	7H	Sealing set
5E	Pin	7K	Repair set, brake pads
6	Spring clip	8	Bleeder valve
7	Metal cover	12	Fixed caliper fastening bolt
7C	Piston	16	Backplate
7E	Piston	19	Bolt

Worksheet for Adjusting Axle Drive

Axle model 737.2 – 747.2
737.3 – 747.3

UNIMOG
MB-trac



Customer

Order No.

Date

Dealer

first registered

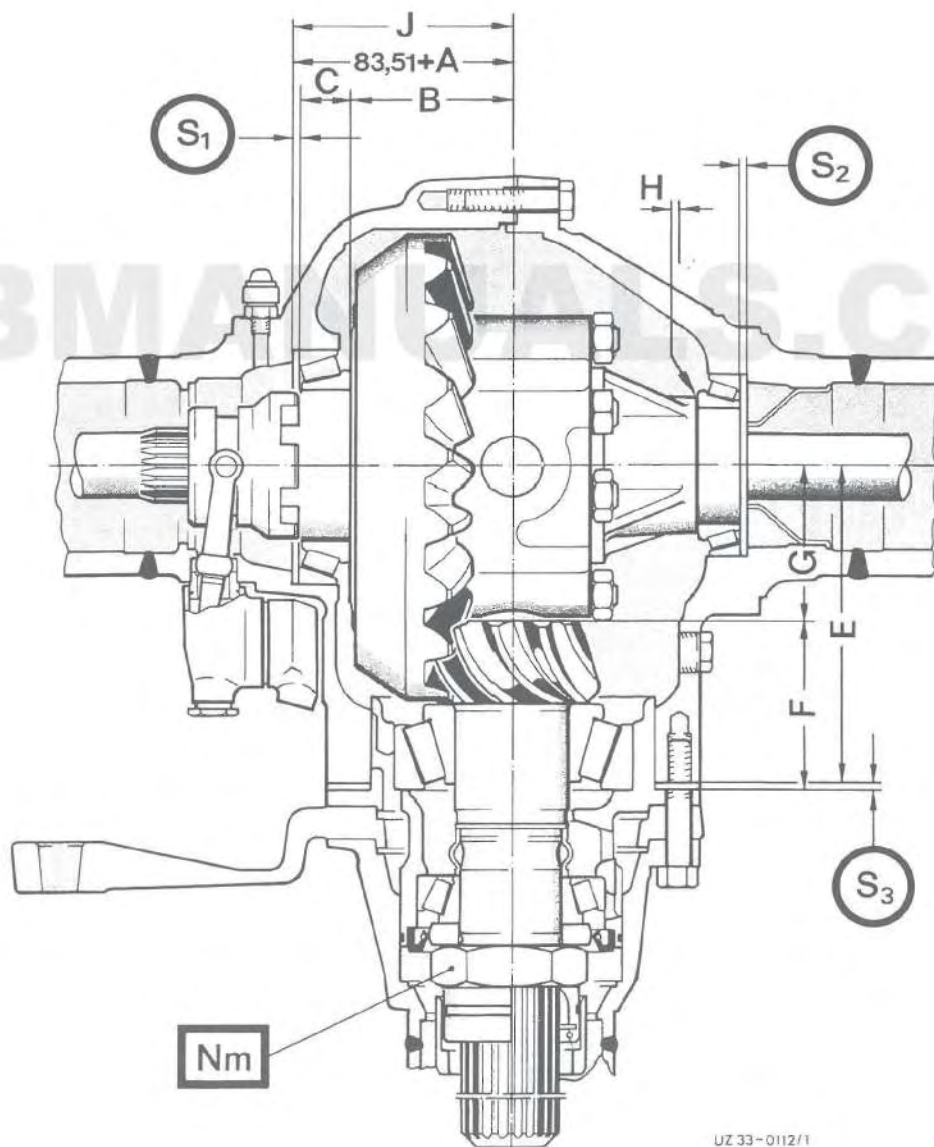
Chassis No.

Axle No.

Operating hours/odometer reading

Last service

carried out by

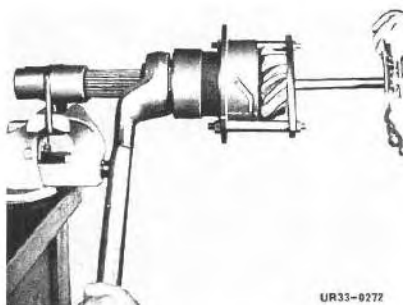


UZ 33-0112/1

33.3 Adjusting Axle Drive

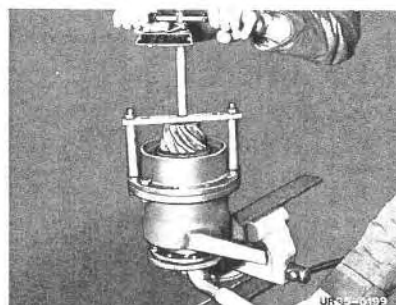
Nm

Adjust breakaway
drive pinion bearing
force for



UR33-0272

Front axle



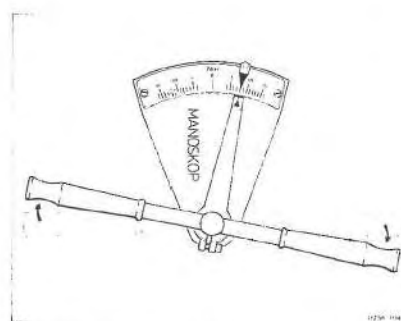
UR33-0199

Rear axle

Measure and adjust preload

- Use new bearings
- Remove preservative from bearing and oil
- Only use new clamping sleeve
- Exactly maintain breakaway force!

Axle model	Breakaway forces
737.2/747.2	6,0 to 6,5 nm
737.30/747.30	7,5 to 8,0 nm
737.31/747.31	6,5 to 7,0 nm



UR33-0041

Use torque wrench

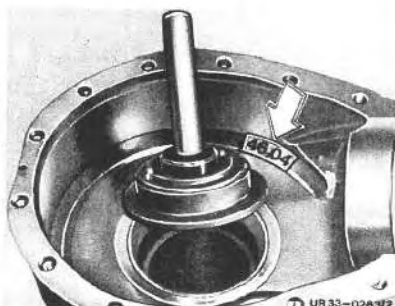


Calculate shim thickness S_1 (large bearing)

Enter dimensions here

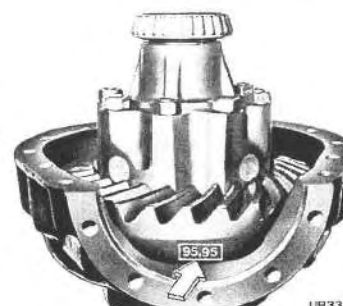
Example

Dimension (const.)	=	83,51	(83,51)
+ Dimension A	=		(46,04)
Dimension J	=	mm	(129,55)



UR33-0283/2

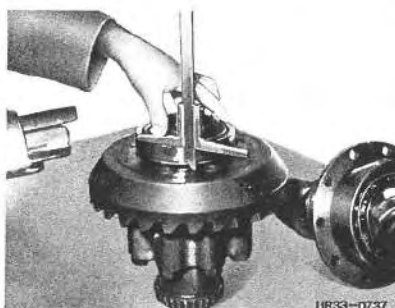
Read off dimension A at housing



UR33-0730/1

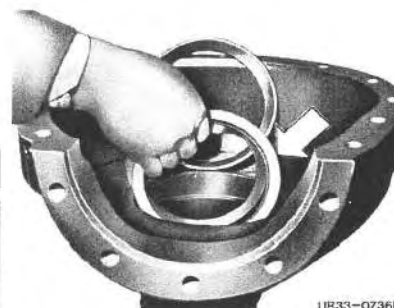
Read off dimension B at ring gear

Dimension A	=	(95,95)
+ Dimension C	=	(32,00)
Dimension B+C	=	mm (127,95)



UR33-0737

Measure dimension C (bearing height) Insert shims S_1 mm



UR33-0736/1

Dimension J	=	(129,55)
- Dimension B+C	=	(127,95)
Shim S_1	=	mm (1,60)



Calculate shim thickness S_2 (small bearing)

- Install outer bearing race **without** shims.
- Mount bearing on differential housing, press in to approx. 4 mm before bearing seat
- Install differential gearing in axle housing.
- Assemble both axle tubes without drive pinion, fasten with some screws (200 Nm) and release once again.
- Measure gap H using feeler gauge, add 0.2 mm preload dimension I .
- One again withdraw outer bearing race and insert shims S_2

Enter dimensions here

Example

	Dimension H =	(1,40)
+ Preload	Dimension I = 0,20	(0,20)
Shim S_2 = mm		(1,60)



UR33-0735/1

Measure dimension H 

Calculate shim thickness S_3 (Flange-drive pinion bearing)

Enter dimensions here

Example

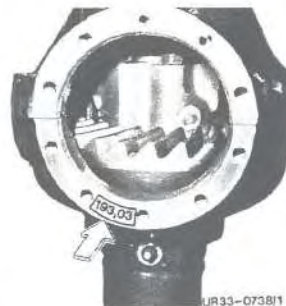
Dimension	F =	(101,60)
+ Dimension	G =	(93,20)
Dimension $F+G$ = mm		(194,80)



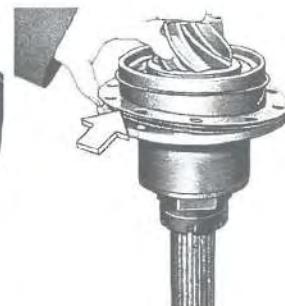
UR33-0728

Measure dimension F 

UR33-0729/1

Read off dimension G 

UR33-0738/1

Read off dimension E at housing

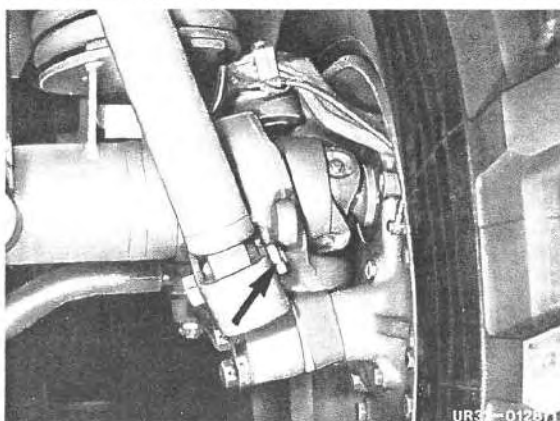
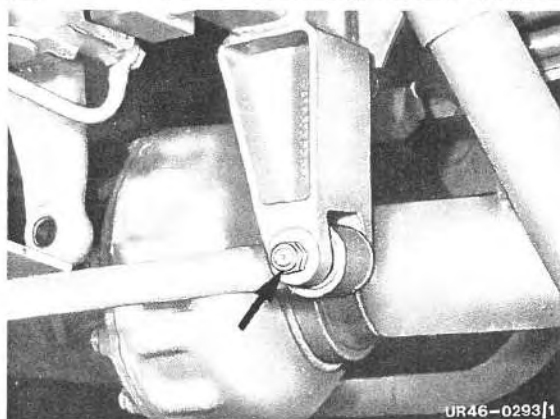
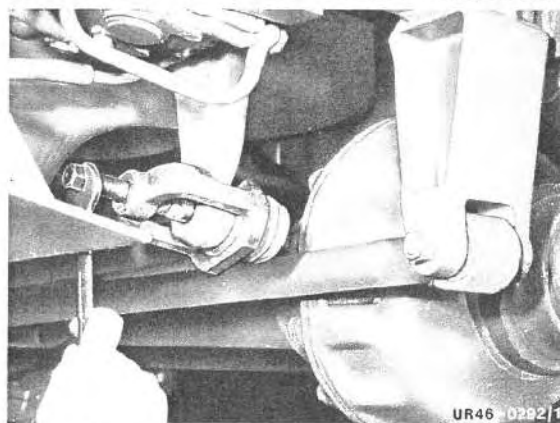
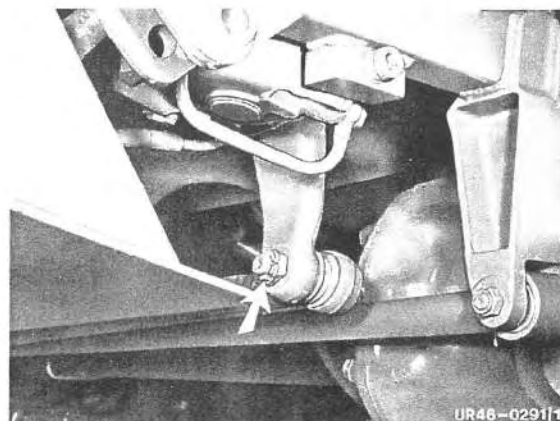
UR33-0741/1

Insert shims S_3 mm.

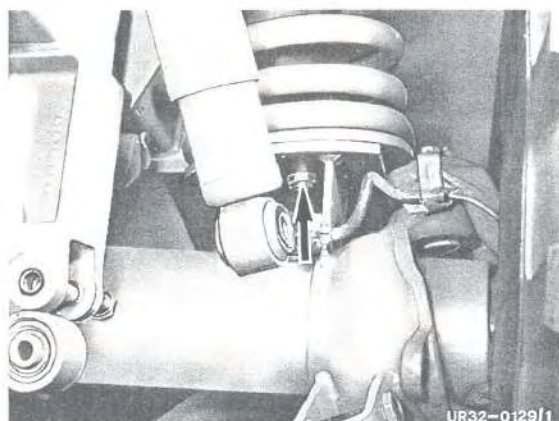
Dimension F+G =	(194,80)
- Dimension E =	(193,03)
Shim (S ₃) = mm	(1,77)

Removal

- 1 Remove spare wheel.
- 2 Release and unscrew castle nut on drag link.
- 3 Using special tool No. 8, draw ball joint off pitman arm.
- 4 Unscrew transverse link.
- 5 Unscrew shock absorber at axle.

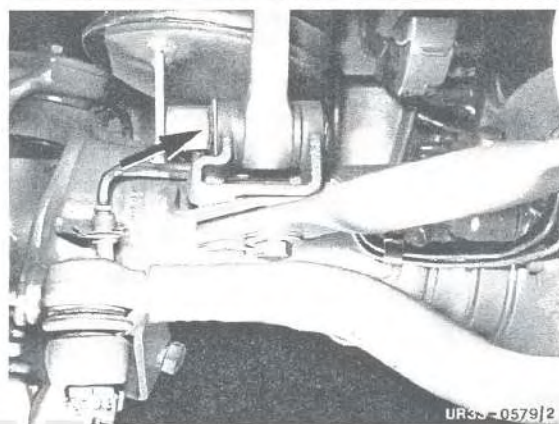


737.2



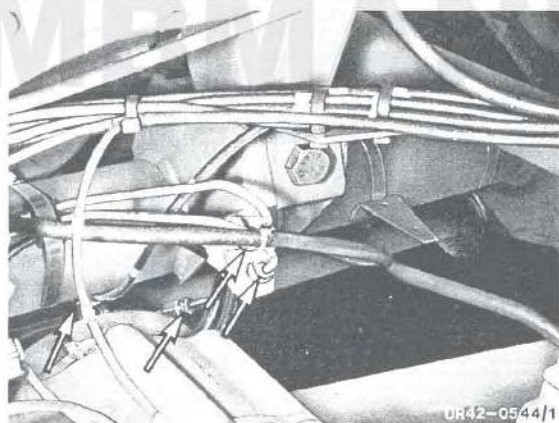
UR32-0129/1

- 6 Unscrew both front springs.



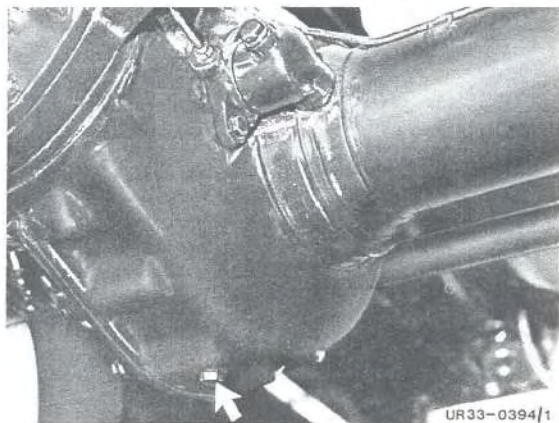
UR33-0579/2

- 7 Unscrew stabilizer at right and left.



UR42-0544/1

- 8 Detach front brake circuit lines, differential lock line and bleeder line. Close ends of brake lines.



UR33-0394/1

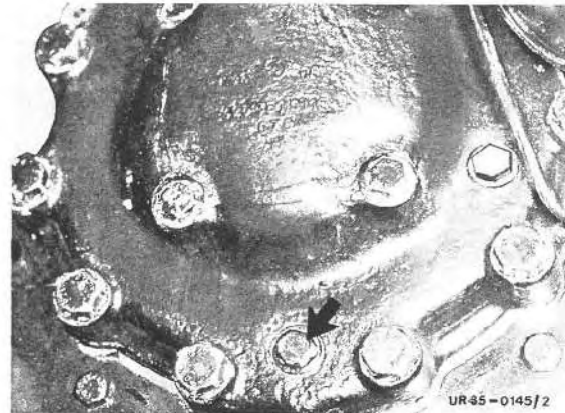
- 9 Drain oil off center axle housing.

Note:

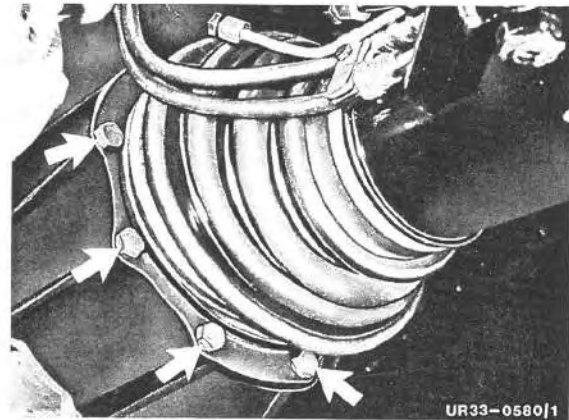
From model	Chassis end No.	Front axle end No.
435.110	070 750	—
.111		

Oil filler and oil drain plug were standardized (hexagon socket M 24 x 1.5).

10 Drain oil off wheel hub drive.



11 Release rubber gaiter and unscrew torque ball casing.



12 Lift up vehicle at front, draw axle forwards and support torque tube.

Note: To raise vehicle, attach chain tackle to both front frame ends.



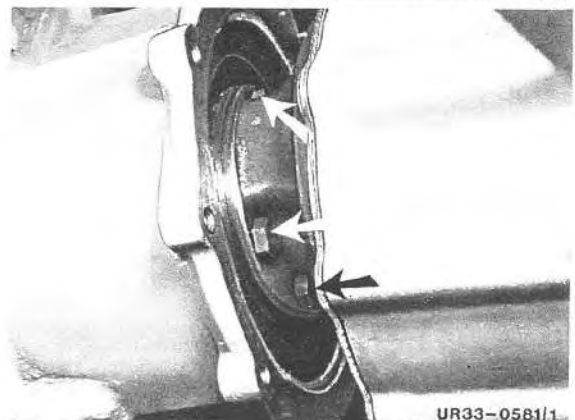
13 Remove front torque ball shell halves.

14 Unscrew propeller shaft from transmission.

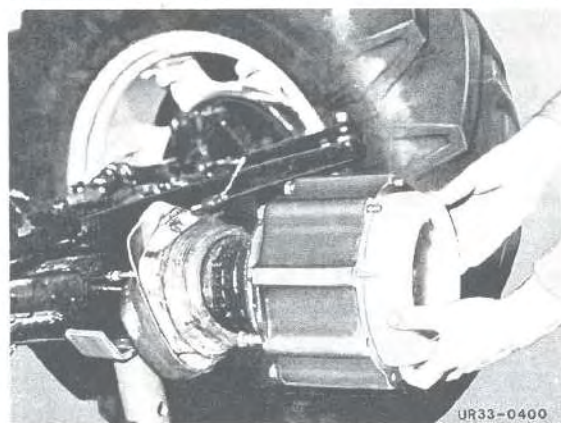
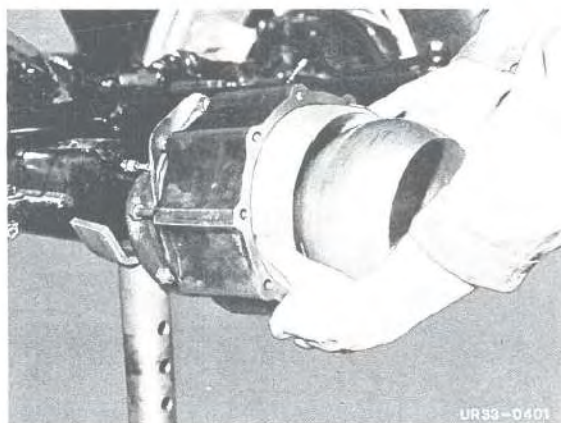
15 Lift vehicle out of springs and draw axle forwards.

16 Deposit the vehicle on trestles.

17 Draw torque ball casing back and remove rear torque ball shell halves.



18 Clean and check torque ball shell halves and torque ball casing, exchanging if necessary.



Preparing for installation

1 Carefully deburr torque ball housing and torque ball shells.

2 Coat rear ball shells with running-in paste No. 3 and install.

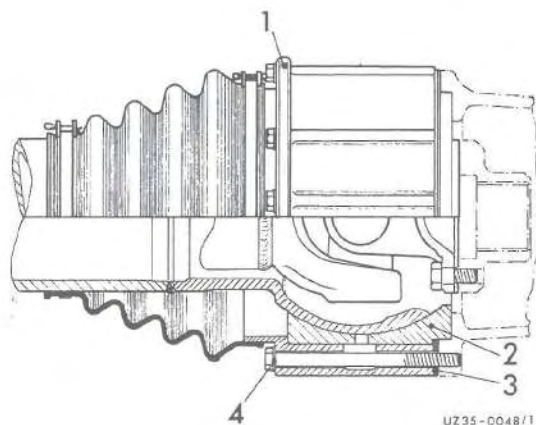
3 Coat front ball shells with running-in paste No. 3 and install.

Note:

From Model	Chassis end No.	Front axle end No.
435.110		
.111	076 650	002 699
.113		
.170		

The shims, the torque ball housing and the torque ball shell were modified as follows:

Item	Designation	Modification
1	Thrust ball housing	Recess in housing eliminated (see ill., Item a)
2	Thrust ball shells front	Overall length reduced from 66 mm to 62 mm.
3	Shims	Change in shape, see ill.
4	Sealing ring A 10 x 16	For screw head/torque ball housing, eliminated.
5	Sealing compound Terostat 56	Used as seal between torque ball housing and transmission flange.

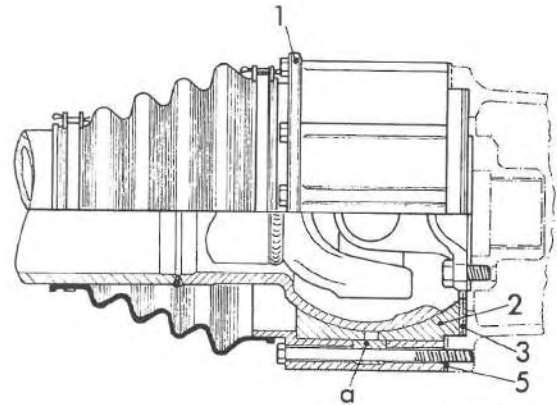


4 Position of shims (3) previously.

Note:

When installing SA 35 973 and SA 35 974 (front or rear axle components with fording ability), the sealing ring previously fitted it not required.

- 5 Position of shims (3), new.

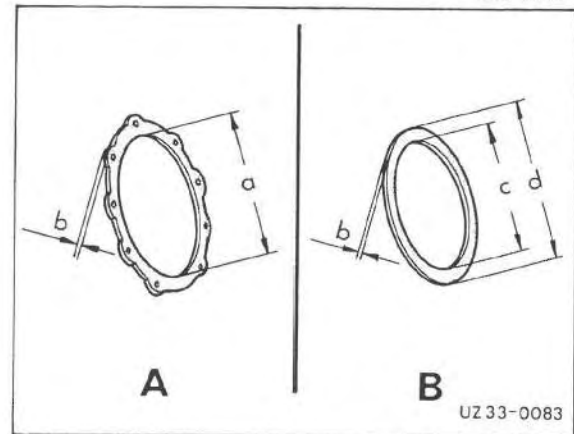


UZ 35-0049/1

- 6 Install shims for measuring.

Note:

- A Shims, previously
- B Shims, new
- a Dia. 205 mm
- b 0.2; 0.5; 1.0 mm
- c Dia. 148 mm
- d Dia. 194 mm

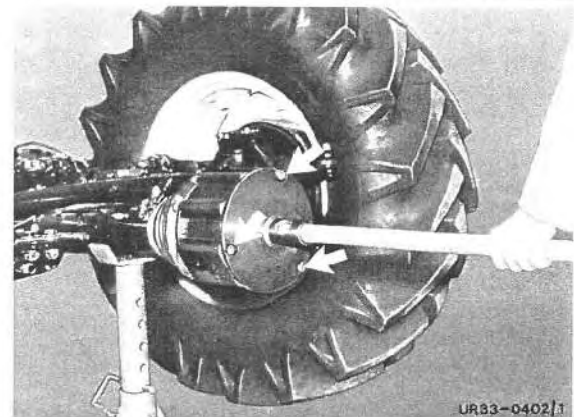


UZ 33-0083

- 7 Screw on special tool No. 17 and tighten fastening bolts **evenly** until a manual force of 200 to 300 N is achieved at the extension tube.

Note: Remove shims if manual force required is inadequate, and add shims if force excessive.

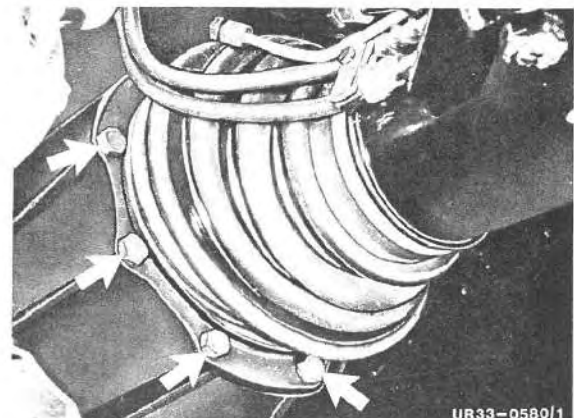
If the manual force required is inadequate when using the new shims, shims must be inserted. If the manual force required is excessive, shims must be removed.



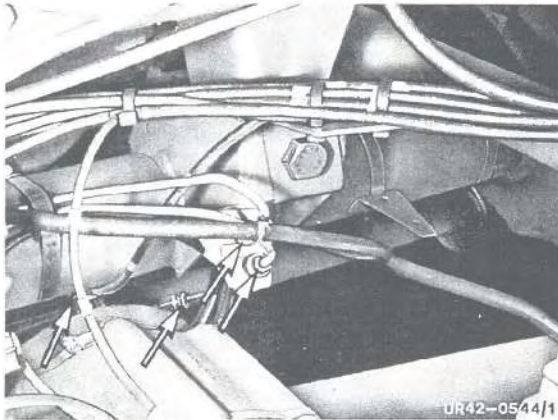
UR83-0402/1

Installation

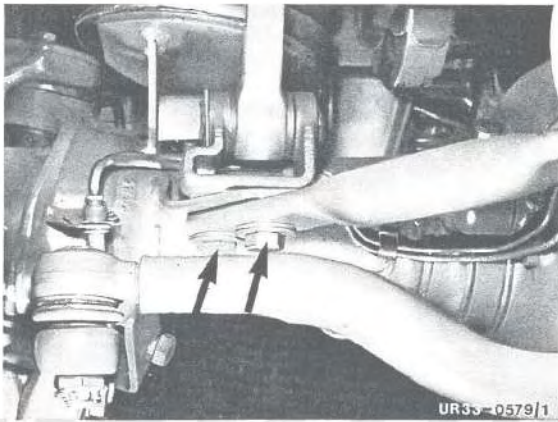
- 1 Raise vehicle and move axle into position.
- 2 Attach propeller shaft to transmission.
- 3 Attach and tighten torque ball casing with shims of calculated thickness.



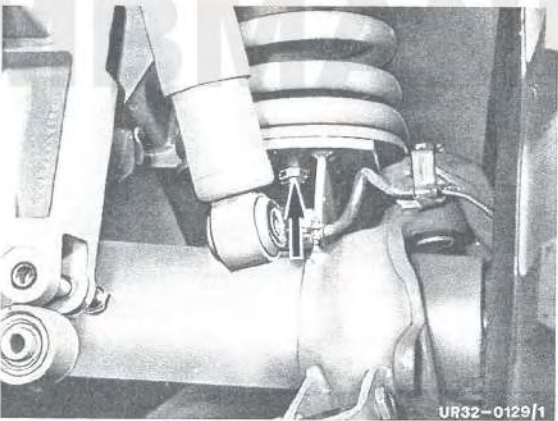
UR33-0580/1



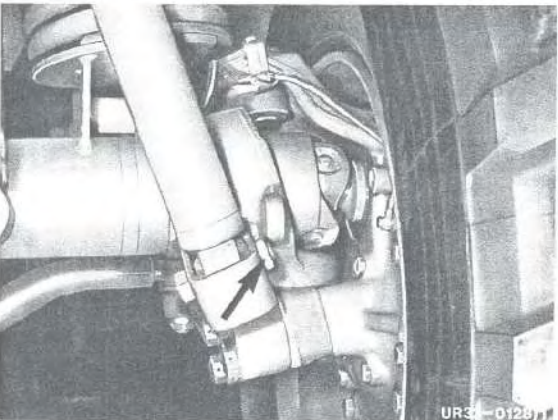
- 4 Attach brake lines for front brake circuits, line for differential lock and bleed line.



- 5 Screw on stabilizer at right and left.

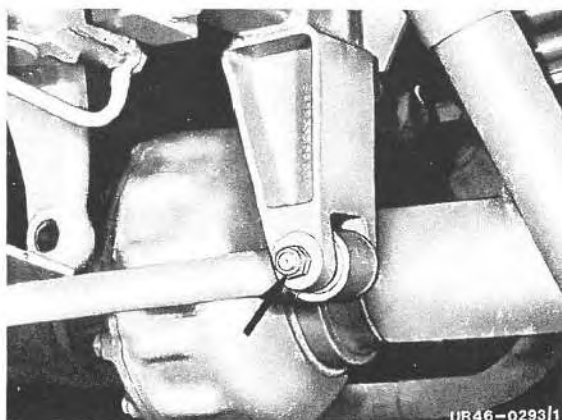


- 6 Screw on bottom bolts at both front springs.



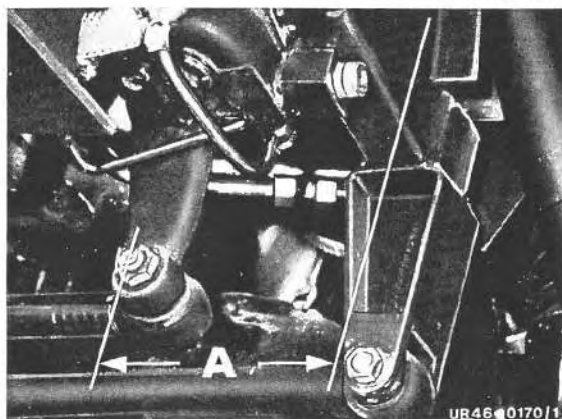
- 7 Attach shock absorber to axle.
Refer to 1.5/1 for tightening torques.

- 8 Screw on transverse link.

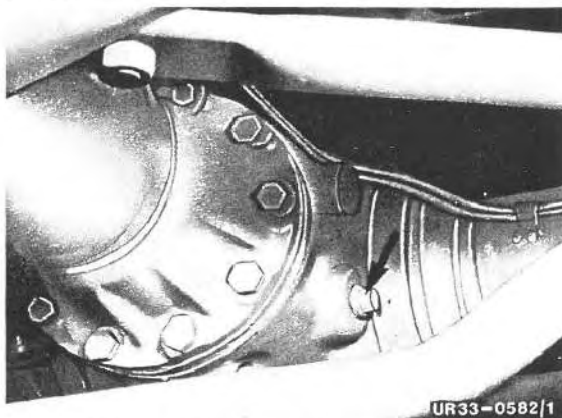


- 9 Attach steering rod in pitman arm, tighten fastening nut and secure.
Refer to 1.5/1 for tightening torques.

Note: Check dim. A from centre pitman arm to outer edge of frame, adjusting if necessary.
Refer to 1.3/1 for adjustment dim.



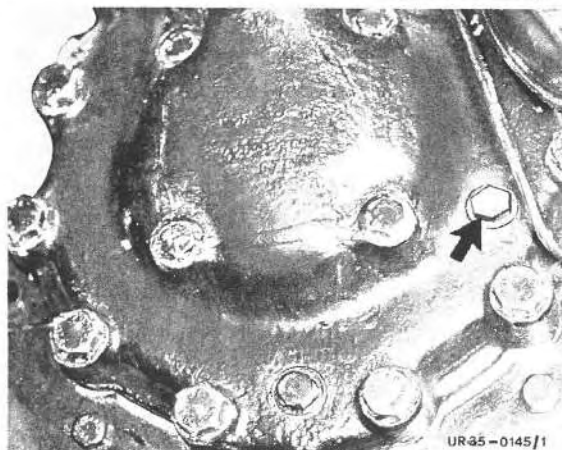
- 10 Fill oil into axle centre housing.
Refer to 1.1/2 for capacity.



- 11 Fill oil into wheel hub drive.
Refer to 1.1/2 for capacity.

- 12 Bleed brake system.

- 13 Check brake system for function and leaks.



Disassembly

1 Clean front axle, clamp in trestle and remove wheels.

2 Unscrew brake lines and line to differential lock and axle bleeder.

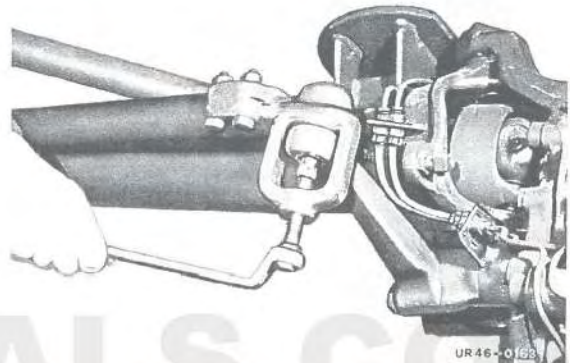
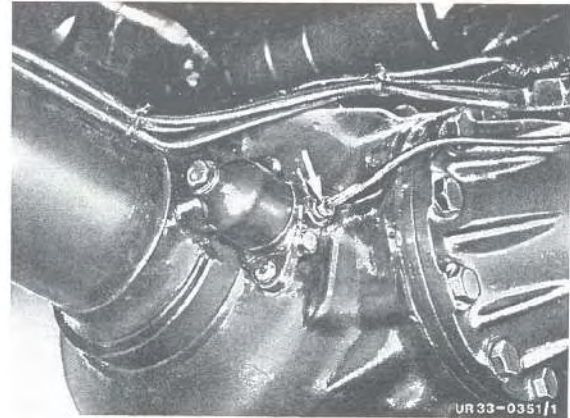
Note:

From Model Chassis end No. Front axle No.
435.111 001 178

Axle bleeder installed on the left in axle tube.

3 Remove drag link and track rod using special tool No. 8.

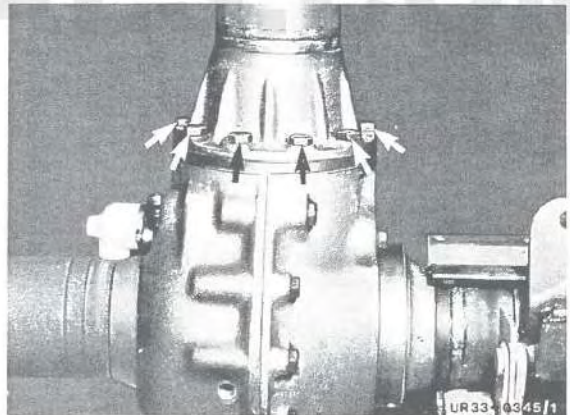
4 Unscrew axle struts.



5 Withdraw propeller shaft from torque tube, unscrewing and removing torque tube.

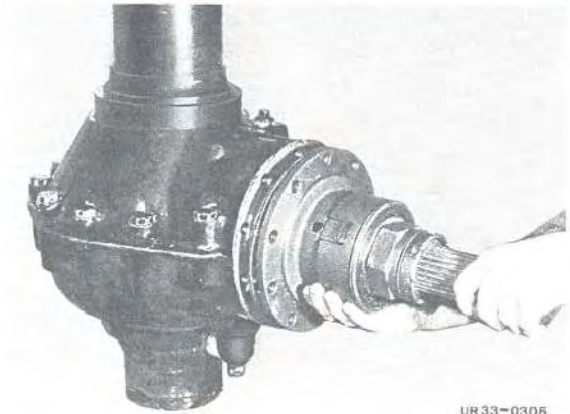
6 Disassemble wheel hub drive, referring to 6.1/1.

7 Turn axle by 90° in trestle.



8 Release connecting bolts of the two axle tubes and remove pinion flange with drive pinion.

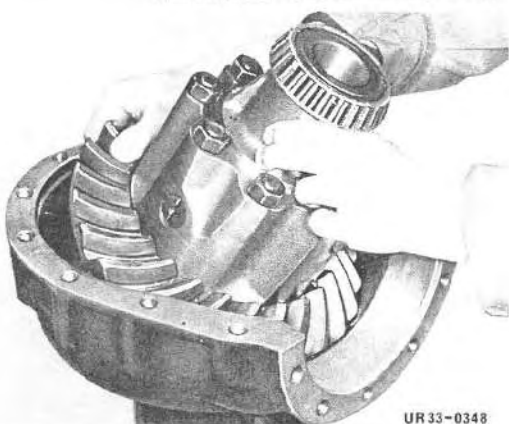
Note: Take care not to lose shims between pinion flange and axle housing.



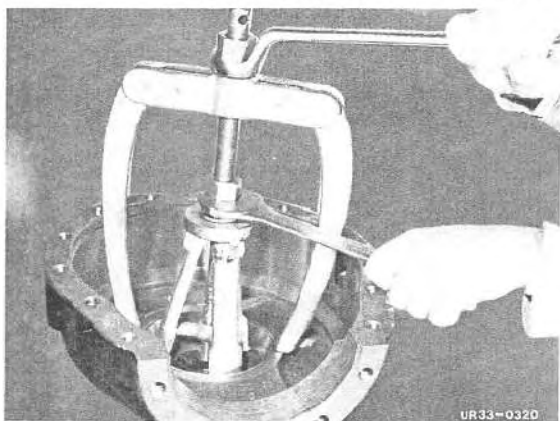
737.2



9 Disconnect axle tube from axle drive.

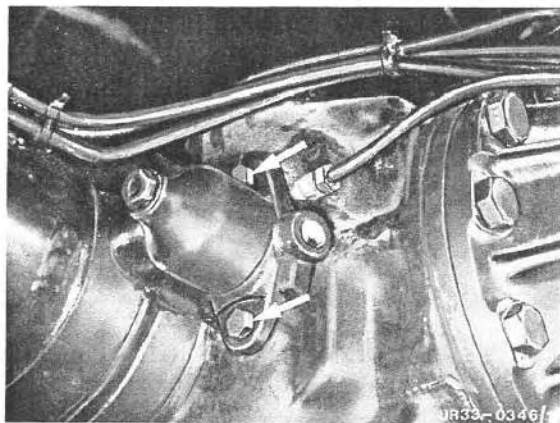


10 Take differential out of axle tube.



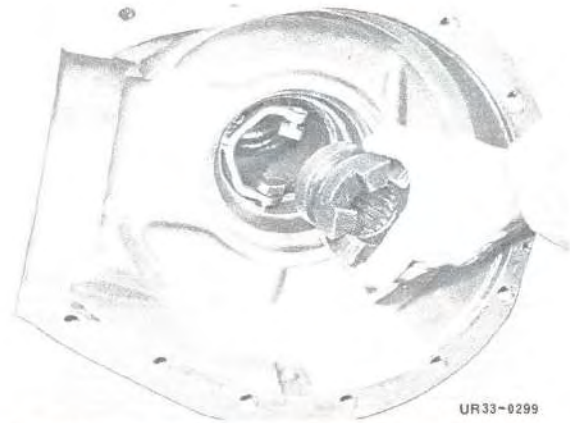
11 Using puller and special tool No. 11, withdraw outer taper roller bearing race from both axle tubes.

12 Turn axle tube by 90° in trestle.



13 Unscrew and remove differential lock shift cylinder.

- 14 Take shift dog out of axle tube.

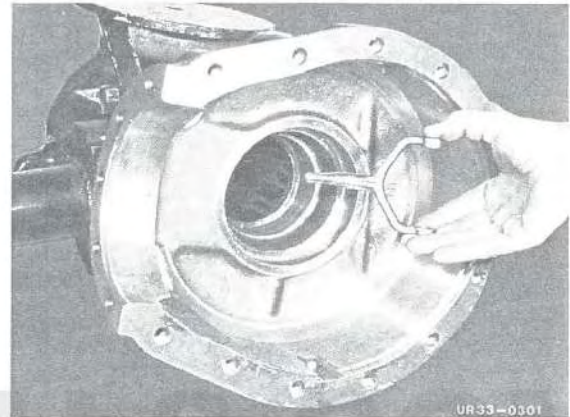


UR33-0299

- 15 Take snap ring off selector fork, remove selector fork from swivel bearing.

- 16 Knock swivel bearing out.

- 17 Clean and check all parts, exchange if necessary.



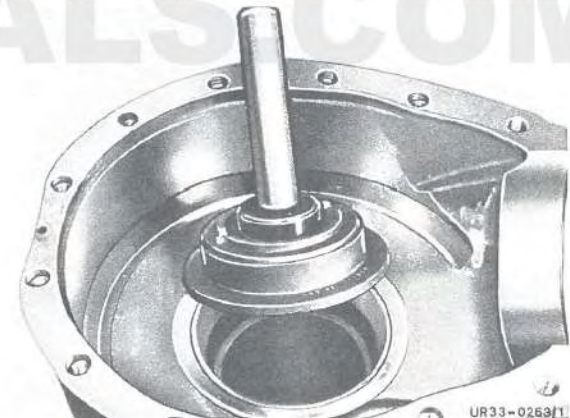
UR33-0301

Assembly

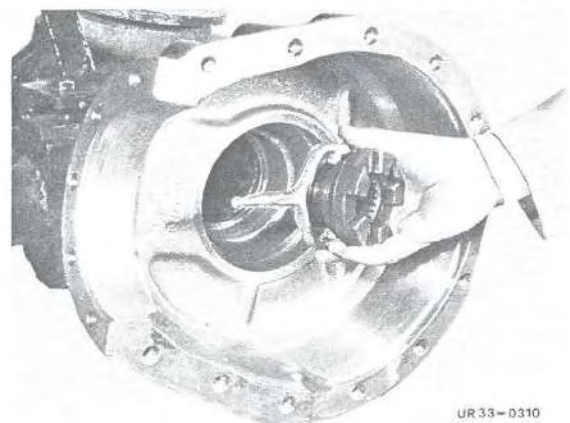
Note: Prior to assembling the axle, be sure to adjust axle drive with reference to the "Work Sheet for Adjusting Axle Drive".

During the operation "Adjustment of Axle Drive" the axle tube (lock side) has already been clamped in the trestle, both outer taper roller bearing races have been driven into the axle tubes with special tool No. 24, and the inner taper roller bearing races have been knocked onto the differential housing together with shims.

- 1 Clamp axle tube horizontally in trestle.
- 2 Install selector fork with sliding blocks and shift dog in axle tube.

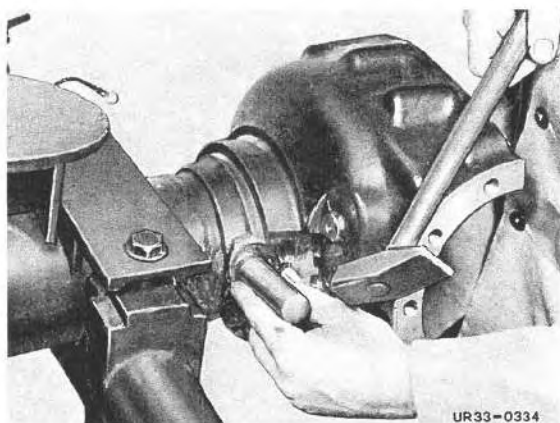


UR33-0263/1

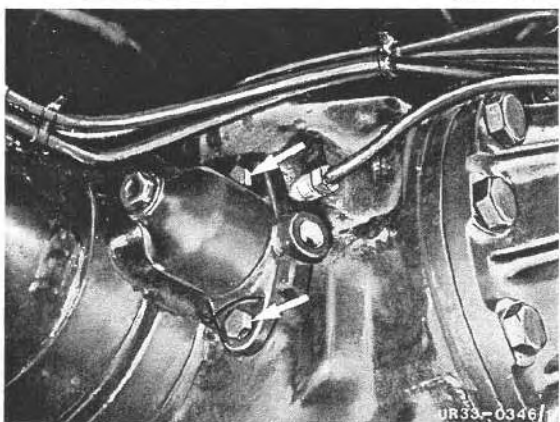


UR33-0310

737.2

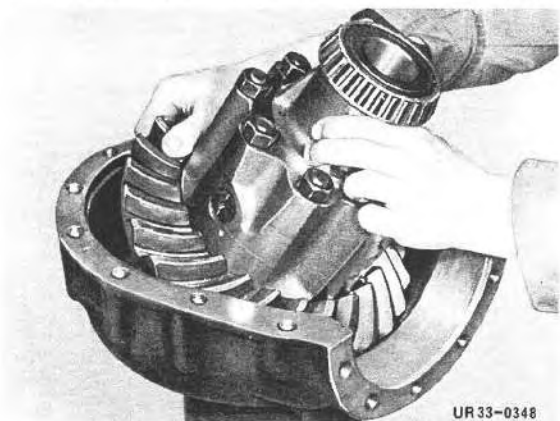


- 3 Install swivel joint with special tool No. 22.

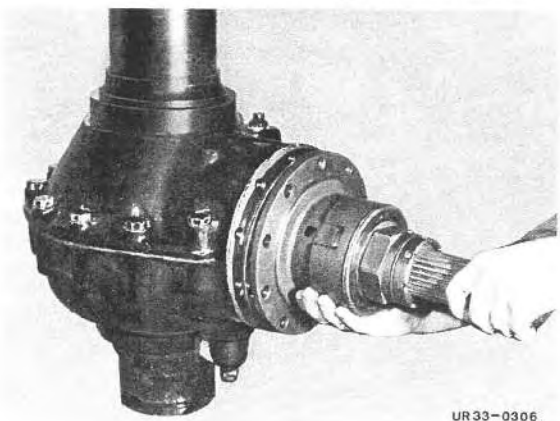


- 4 Attach shift cylinder to differential lock.

- 5 Turn axle tube by 90° in trestle.



- 6 Install differential in axle tube at lock side.

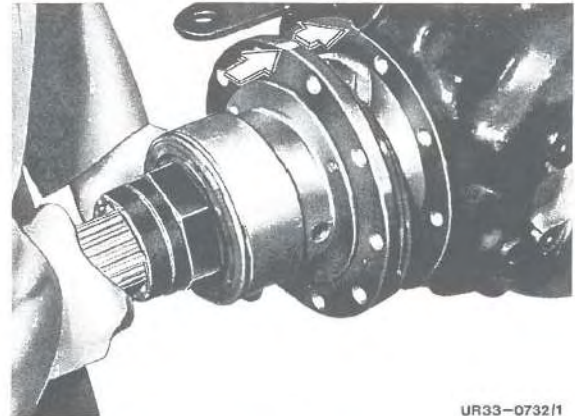


- 7 Coat both axle tubes with sealant No. 1 and then bolt together.

Note: To center axle housing correctly, insert pinion flange into housing.

8 Coat spline on drive pinion bearing with running-in paste No. 7 and install with the shim rings determined on the dimension sheet and O-ring in the axle housing.

Note: Notch and oil duct facing upwards.



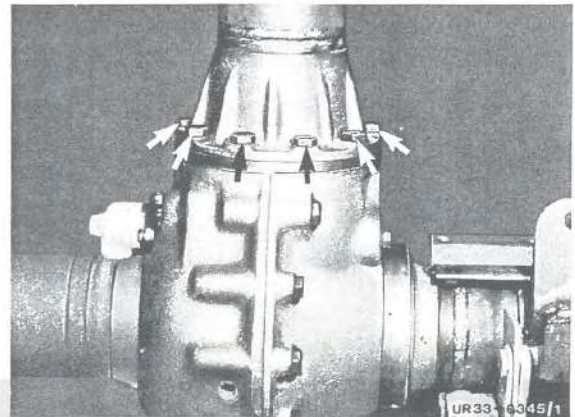
UR33-0732/1

9 Coat torque tube flange with sealing compound No. 1 and bolt torque tube to pinion flange.

10 Turn axle in assembly base 90°.

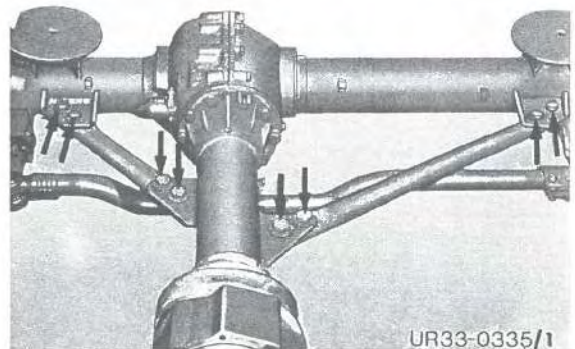
11 Clean spline and cavity of propeller shaft, fill cavity behind the profile with lubricating grease No. 2 (50 gr.) and install propeller shaft.

12 Assemble wheel hub drive refer to 6.1/1, coat with sealing compound No. 1 and install.



UR33-0345/1

13 Coat fastening bolts with locking compound No. 4 and attach axle struts.



UR33-0335/1

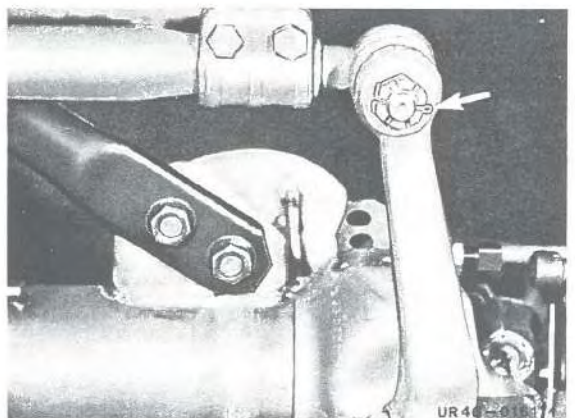
14 Attach drag link and track rod, tighten castle nut and secure with cotter pin.

15 Attach brake, differential lock and axle breather lines.

16 Remove axle from assembly block.

17 Attach road wheels and tighten.
For tightening torques, refer to 1.5/1.

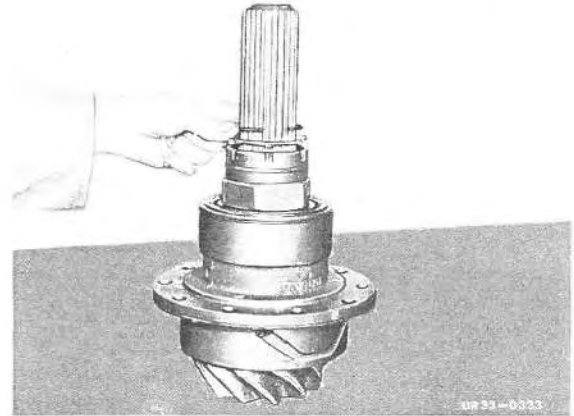
Note: Retighten wheel nuts after 50 km.



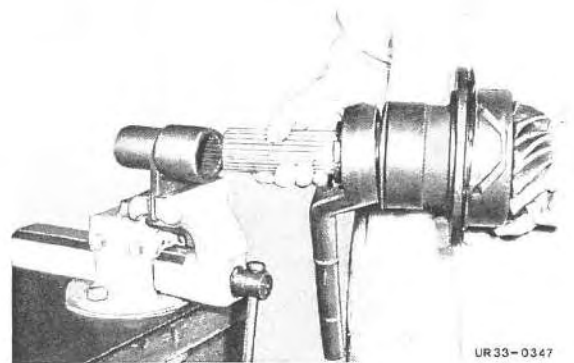
UR40-0181/1

Disassembly

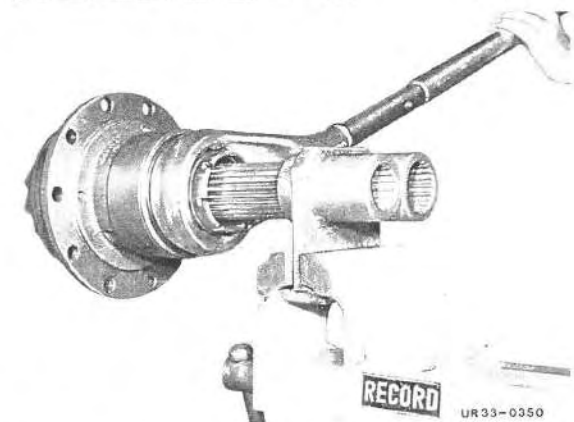
1 Remove lock washer and serrated washer from hexagon nut.



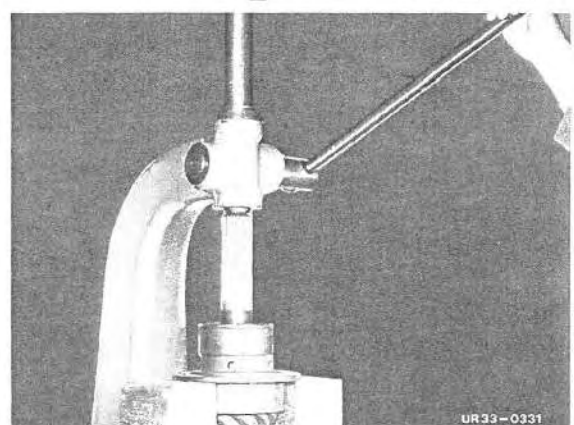
2 Clamp drive pinion in vise, using special tool No. 20.



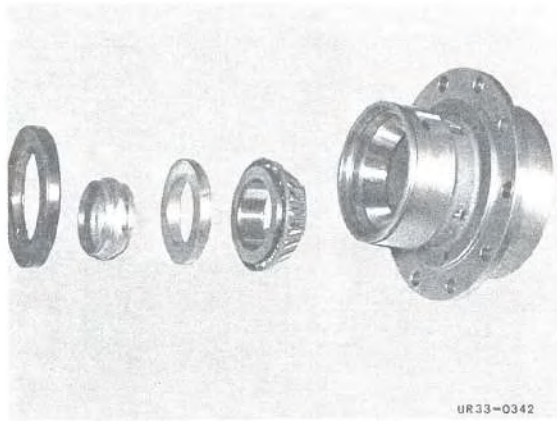
3 Unscrew hexagon nut.



4 Force drive pinion out.

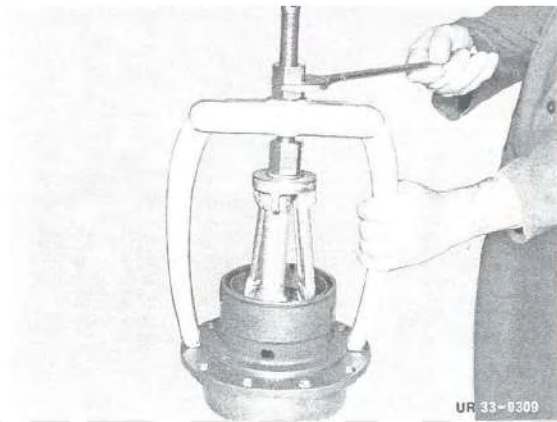


737.2



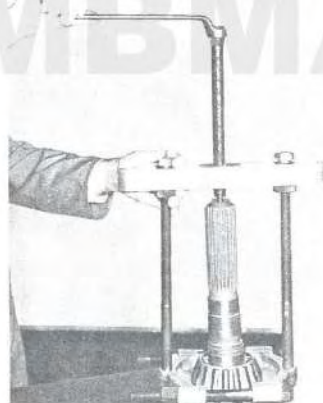
UR33-0342

5 Remove race, shaft seal, taper roller bearing and clamping sleeve from pinion flange.



UR 33-0309

6 Draw outer races of taper roller bearing off pinion flange, using special tool No. 11.

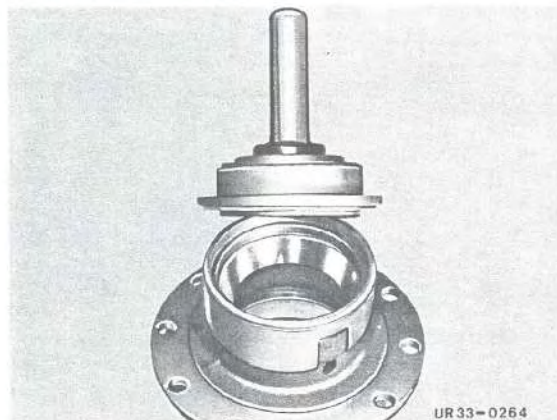


UR33-0332

7 Draw taper roller bearing and inner race off drive pinion.

Note: Clean and check all parts, exchanging if necessary.

Prior to installing taper roller bearings, wash out thoroughly and insert with SAE 90 oil.



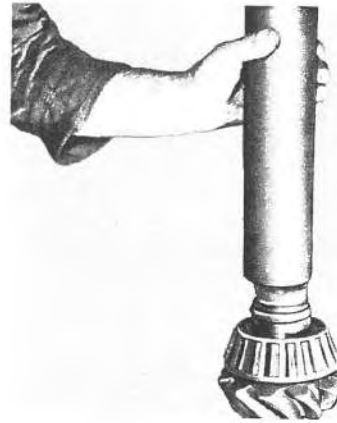
UR33-0264

Important: If drive pinion is replaced, it is always to be exchanged **together** with ring gear. Make sure that serial numbers are identical.

Assembly

1 Force outer race of taper roller bearing into pinion flange, using special tool No. 24.

- 2 Press tapered roller bearing onto drive pinion using special tool No. 21.
- 3 Introduce drive pinion into pinion flange.

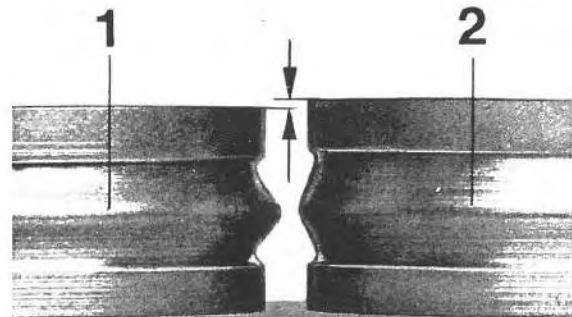


UR 33-0265

- 4 Use new clamping sleeve.

Note: Use only a **new clamping sleeve (2)** since the old clamping sleeve (1) is compressed during initial installation.

- 5 Coat tapered roller bearings with lubricating grease No. 2 and press into bearing seat using special tool No. 21 so that bearing has a 1 to 2 mm clearance.



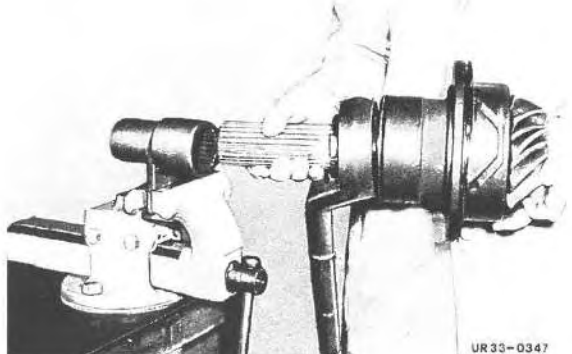
UR 33-0009

- 6 Coat shaft sealing ring with lubricating grease No. 2 and install using special tools No. 25 and No. 21.
- 7 Install bearing race, screw on hexagon nut.



UR 33-0289

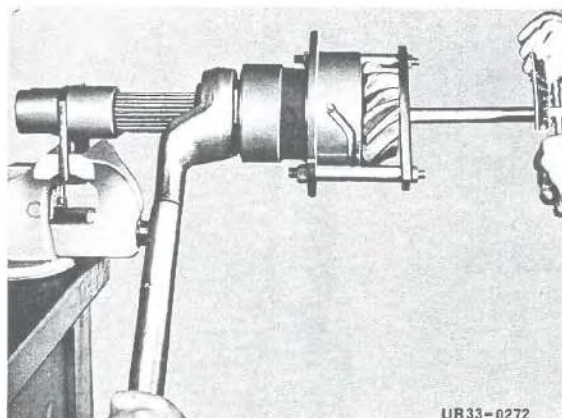
- 8 Clamp drive pinion together with special tool No. 20 in vise and tighten hexagon nut until all the bearing play has been taken up.



UR 33-0347

33.3 Disassembly and Assembly of Drive Pinion Bearing

737.2

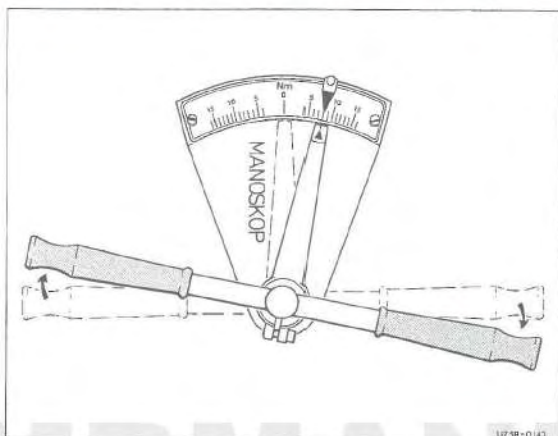


Adjusting Drive Pinion

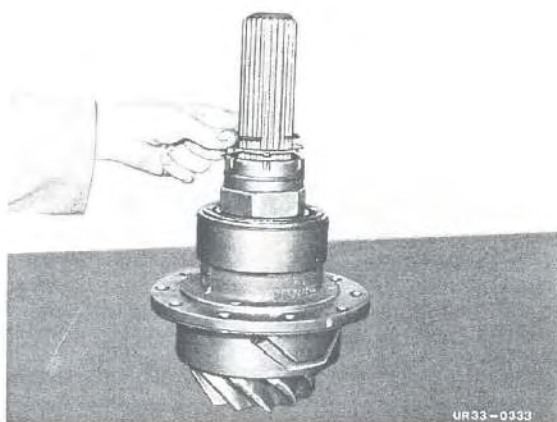
- 1 Mount special tools No. 18 and No. 19 on pinion flange.
Refer to worksheet 1.7/1.
- 2 Screw on large nut and tighten until all perceptible play of the flange is eliminated.

Note: Tighten nut slowly, while constantly checking the breakaway force to ensure the value is not exceeded.

Prior to checking with the torque wrench turn the flange several times by hand.



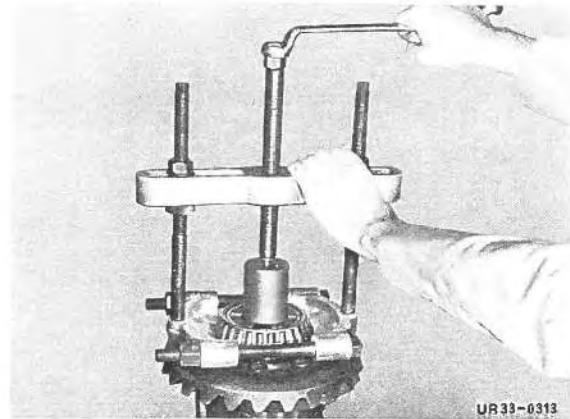
The preload (breakaway force) is correctly adjusted if the flange begins to turn at a value from 6,0 to 6,5 Nm. If this value is exceeded, a new clamping sleeve must be installed once again.



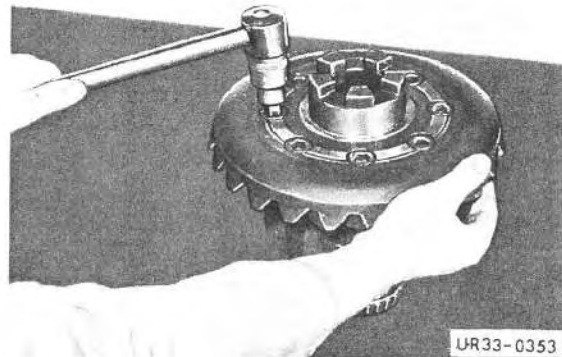
- 3 Install serrated washer, lock washer and sealing ring in hexagon nut.

Disassembly

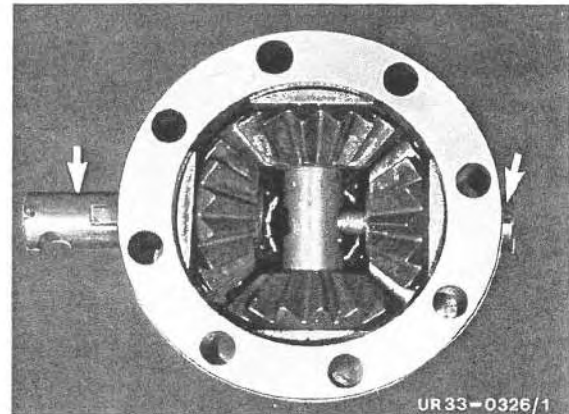
- 1 Remove front axle, referring to 2.1/1.
- 2 Disassembly front axle, referring to 3.1/1.
- 3 Remove both inner taper roller bearing races from differential, using puller.



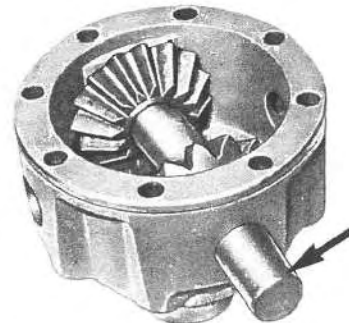
- 4 Unscrew and remove ring gear from differential housing, using special tool No. 3.



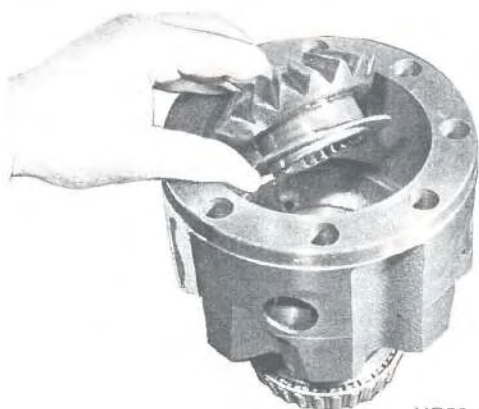
- 5 Remove thrust washer and side gear.
- 6 Remove both short differential pinion shafts.
- 7 Take out thrust segments and differential pinions.



- 8 Remove long differential pinion shaft.
- 9 Take out thrust segments and differential pinions.



737.2



UR33-0330

10 Take side gear and thrust washer out of differential housing.

11 Clean and check all parts, exchange if necessary.

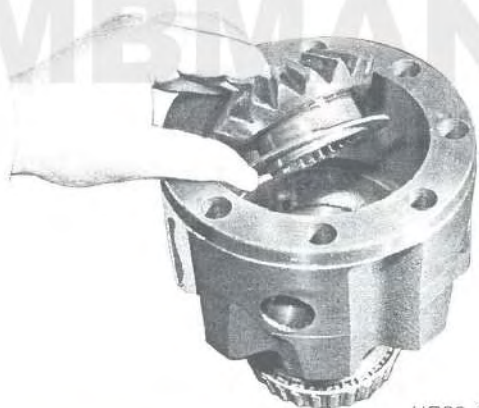


UR33-0008

Assembly

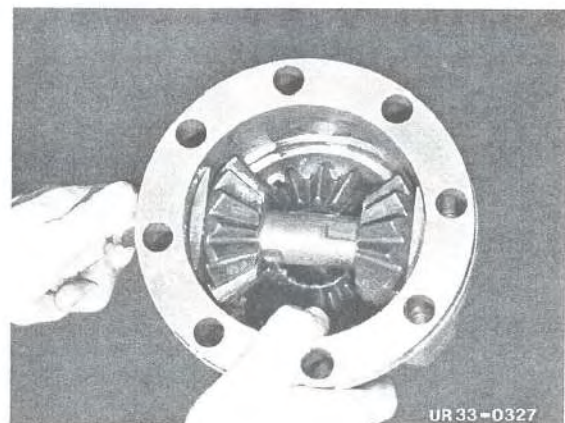
Note: Ring gear must be exchanged together with drive pinion. Make sure serial numbers are identical.

1 Coat all differential shafts and thrust segments with paste No. 3.



UR33-0330

2 Install thrust washer and side gear in differential housing.

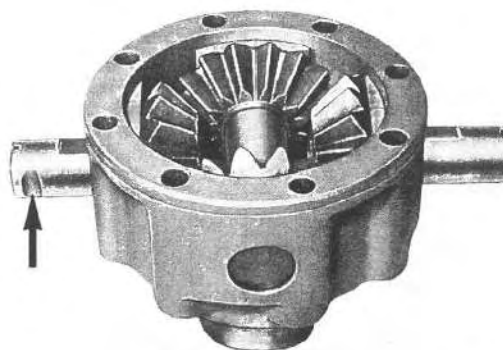


UR33-0327

3 Install thrust segments, differential pinions and differential shafts.

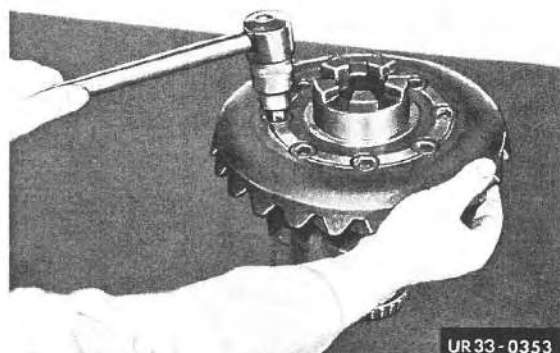
Note: Note recesses in differential pinion shafts for bolts.

4 Introduce thrust washer and side gear in differential housing.



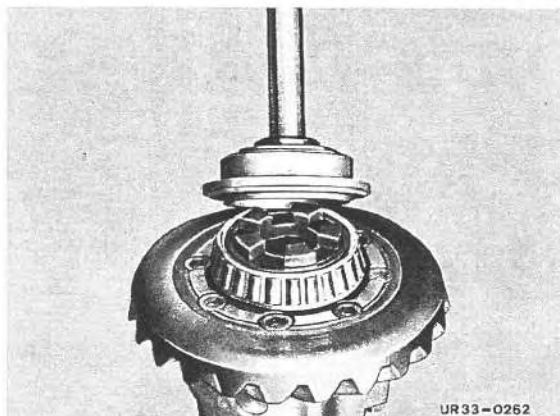
UR33-0473/1

5 Using special tool No. 3, bolt differential housing and ring gear together and torque. Tightening torques, referring to 1.5/1.



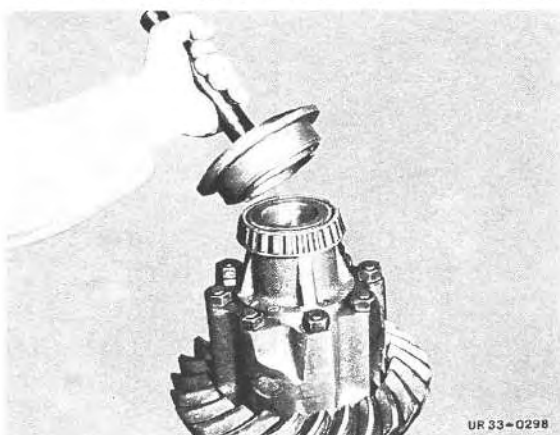
UR33-0353

6 Using special tool No. 24, force large inner race of taper roller bearing onto differential housing and coat with fluid grease No. 6.



UR33-0262

7 Using special tool No. 24, force small inner race to taper roller bearing onto differential housing and coat with fluid grease No. 6.



UR 33-0298

8 Assembly front axle, referring to 3.1/1.